

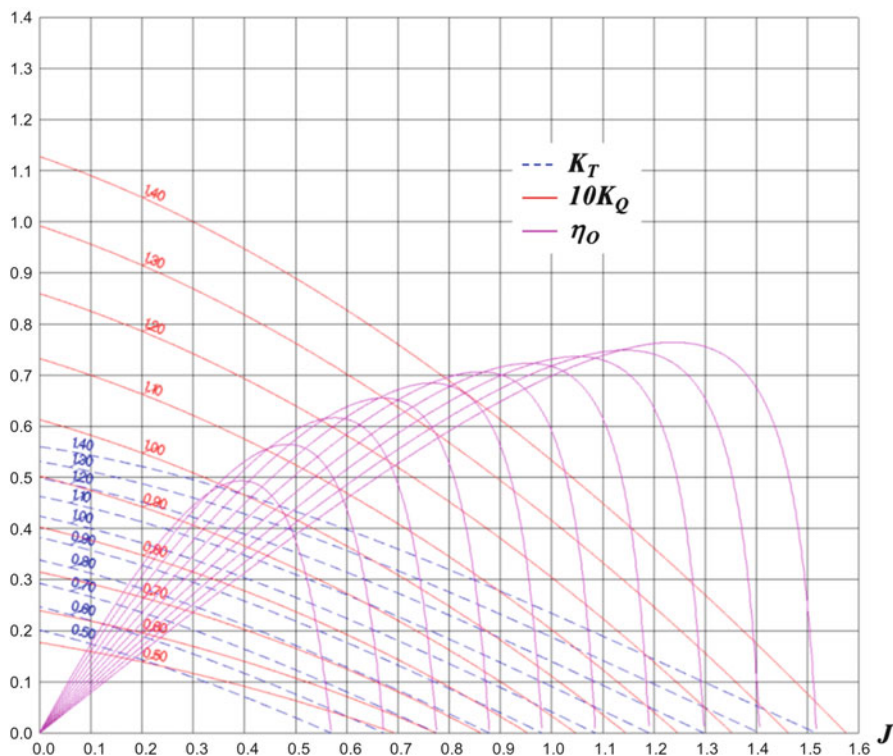


Fig. 7.6 POW diagrams of the Wageningen B-series propellers for $A_E/A_O = 0.55$ and $z = 4$

procedure includes five major stages from Stage 1 to Stage 5. The detailed explanation of each stage is described below.

7.4.1 Stage 1: Assumption of Propeller Diameter

After the total resistance of design ship is estimated, the engine power from the EHP to MCR can be predicted, as described in Sect. 5.6. Then, the delivered power to the propeller (DHP) from the main engine can be assumed with the engine speed. At the early design stage, the engine speed of design ship can be assumed as same as that of parent ship.

Next, the propeller diameter (D_P) of design ship is assumed in Stage 1. There are two methods to estimate the propeller diameter of design ship. The first method is to use the propeller diameter of parent ship as that of design ship, as the following equation.

Table 7.1 Coefficients of the K_T and K_Q polynomials of the Wageningen B-series propellers for $R_n = 2 \times 10^6$

K_T					K_Q				
	s	t	u	v		s	t	u	v
$C_{s,t,u,v}$	(J)	(P_i/D_P)	(A_E/A_O)	(z)	$C_{s,t,u,v}$	(J)	(P_i/D_P)	(A_E/A_O)	(z)
+0.00880496	0	0	0	0	+0.00379368	0	0	0	0
-0.204554	1	0	0	0	+0.00886523	2	0	0	0
+0.166351	0	1	0	0	-0.032241	1	1	0	0
+0.158114	0	2	0	0	+0.00344778	0	2	0	0
-0.147581	2	0	1	0	-0.0408811	0	1	1	0
-0.481497	1	1	1	0	-0.108009	1	1	1	0
+0.415437	0	2	1	0	-0.0885381	2	1	1	0
+0.0144043	0	0	0	1	+0.188561	0	2	1	0
-0.0530054	2	0	0	1	-0.00370871	1	0	0	1
+0.0143481	0	1	0	1	+0.00513696	0	1	0	1
+0.0606826	1	1	0	1	+0.0209449	1	1	0	1
-0.0125894	0	0	1	1	+0.00474319	2	1	0	1
+0.0109689	1	0	1	1	-0.00723408	2	0	1	1
-0.133698	0	3	0	0	+0.00438388	1	1	1	1
+0.00638407	0	6	0	0	-0.0269403	0	2	1	1
-0.00132718	2	6	0	0	+0.0558082	3	0	1	0
+0.168496	3	0	1	0	+0.0161886	0	3	1	0
-0.0507214	0	0	2	0	+0.00318086	1	3	1	0
+0.0854559	2	0	2	0	+0.015896	0	0	2	0
-0.0504475	3	0	2	0	+0.0471729	1	0	2	0
+0.010465	1	6	2	0	+0.0196283	3	0	2	0
-0.00648272	2	6	2	0	-0.0502782	0	1	2	0
-0.00841728	0	3	0	1	-0.030055	3	1	2	0
+0.0168424	1	3	0	1	+0.0417122	2	2	2	0
-0.00102296	3	3	0	1	-0.0397722	0	3	2	0
-0.0317791	0	3	1	1	-0.00350024	0	6	2	0
+0.018604	1	0	2	1	-0.0106854	3	0	0	1
-0.00410798	0	2	2	1	+0.00110903	3	3	0	1
-0.000606848	0	0	0	2	-0.000313912	0	6	0	1
-0.0049819	1	0	0	2	+0.0035985	3	0	1	1
+0.0025983	2	0	0	2	-0.00142121	0	6	1	1
-0.000560528	3	0	0	2	-0.00383637	1	0	2	1
-0.00163652	1	2	0	2	+0.0126803	0	2	2	1
-0.000328787	1	6	0	2	-0.00318278	2	3	2	1
+0.000116502	2	6	0	2	+0.00334268	0	6	2	1
+0.000690904	0	0	1	2	-0.00183491	1	1	0	2
+0.00421749	0	3	1	2	+0.000112451	3	2	0	2
+0.0000565229	3	6	1	2	-0.0000297228	3	6	0	2
-0.00146564	0	3	2	2	+0.000269551	1	0	1	2

(continued)

Table 7.1 (continued)

K_T					K_Q				
	s	t	u	v		s	t	u	v
$C_{s,t,u,v}$	(J)	(P_i/D_P)	(A_E/A_O)	(z)	$C_{s,t,u,v}$	(J)	(P_i/D_P)	(A_E/A_O)	(z)
					+0.00083265	2	0	1	2
					+0.00155334	0	2	1	2
					+0.000302683	0	6	1	2
					−0.0001843	0	0	2	2
					−0.000425399	0	3	2	2
					+0.0000869243	3	3	2	2
					−0.0004659	0	6	2	2
					+0.0000554194	1	6	2	2

$$D_P = D_{P,P} \quad (7.16)$$

where, $D_{P,P}$ is the propeller diameter of parent ship.

The second method is to use an empirical formula if the propeller diameter of parent ship is not available, as the following formula.

$$D_P = 15.4 \cdot c_1 \cdot \left(\frac{MCR_P}{n_{MCR,P}^3} \right)^{0.2} \quad (7.17)$$

where, MCR_P and $n_{MCR,P}$ is MCR and engine speed at MCR of the parent ship, respectively. And c_1 is 1.05 for the propeller with four blades and 1.0 for the propeller with five blades.

7.4.2 Stage 2: Determination of Engine Power and Speed, and Propeller Pitch for Maximization of η_0

In Stage 2, the engine power, the engine speed, and the propeller pitch which maximizes the propeller efficiency with the assumed propeller diameter from Stage 1 are determined. The problem for determining them can be summarized in Table 7.2.

When we solve this problem, we can use two governing equations. The first one is an equation from the relation between the main engine and the propeller, called **torque equation**. That is, this equation says that *the propeller should absorb the torque (Q_P) delivered from the main engine (Q_E)*. This equation of $Q_E = Q_P$ can be stated as the following equation.

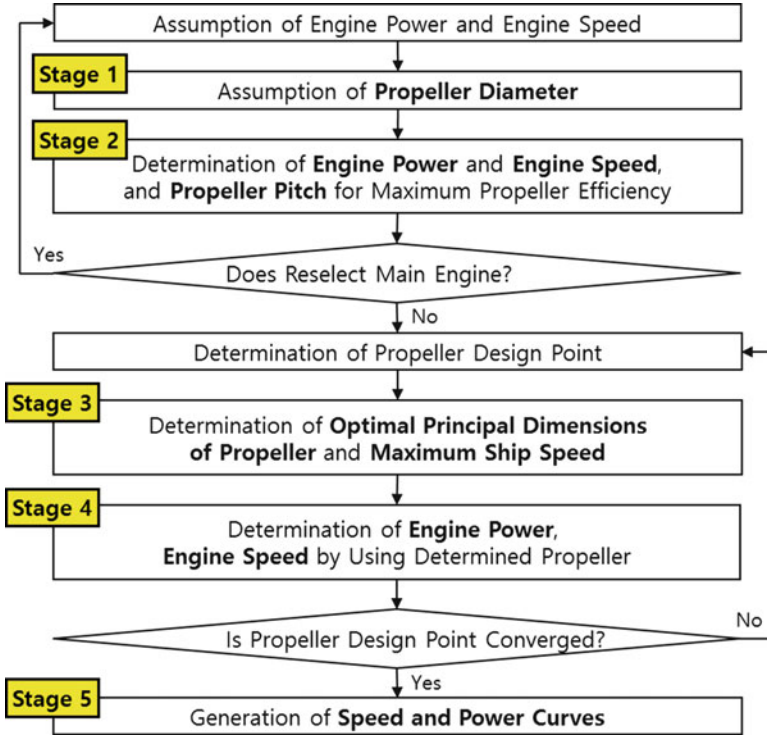


Fig. 7.7 Overall procedure for the determination of principal dimensions of propeller

Table 7.2 Problem definition at Stage 2

	Item	Unit	Description
Given (Assumed)	D_P	m	Propeller diameter
	A_E/A_O	—	Expanded area ratio (EAR)
	z	—	Number of blades
	V	m/s	Ship speed (i.e., Service speed V_S)
	R_T	kN	Resistance in the calm water for given ship speed
Find	P_i	m	Propeller pitch
	P_E	kW	Delivered power to propeller from main engine (= $DHP \cdot \eta_R$)
	n_E	1/s	Engine speed (= Propeller speed, n_P)

$$\frac{P_E}{2\pi \cdot n_E} = \rho \cdot n_P^2 \cdot D_P^5 \cdot K_Q \quad (7.18)$$

where, $n_E = n_P$.

In this equation, the left side means the torque delivered from the main engine (Q_E) and the right side means the torque which the propeller can absorb (Q_P).

The second equation is an equation from the relation between the ship and the propeller, called the **thrust force equation**. That is, the equation says that *the propeller should produce the required thrust force at a given speed*. This equation of $T_S = T_P$ can be stated as below.

$$\frac{R_T}{1-t} = \rho \cdot n_P^2 \cdot D_P^4 \cdot K_T \quad (7.19)$$

In this equation, the left side means the thrust force which is required to propel the ship for the given speed (T_S) and the right side means the thrust force produced by the propeller (T_P).

The problem of Stage 2 has three unknowns (P_i , P_E , and n_E) and two equations. Thus, this problem is a kind of an **indeterminate problem** since *the number of unknowns is greater than the number of equations*. The indeterminate problem has many solutions. If we assume the problem as an **optimization problem** by introducing an **objective function**, we can get the best solution which optimizes the objective function. Here, the objective function is *a criterion to compare different solutions and to determine the best solution*.

Since this problem is relatively simple, we can *solve it by manually*. First, from Eq. (7.19) the thrust coefficient (K_T) can be expressed as a quadratic function with respect to the advance ratio (J).

That is, the thrust force equation in Eq. (7.19) can be rearranged as the following equation.

$$K_T = \frac{R_T}{(1-t) \cdot \rho \cdot D_P^4} \cdot \frac{1}{n_P^2} \quad (7.20)$$

The advance ratio can be expressed as the following equation.

$$n_P = \frac{V_A}{J \cdot D_P} \quad (7.21)$$

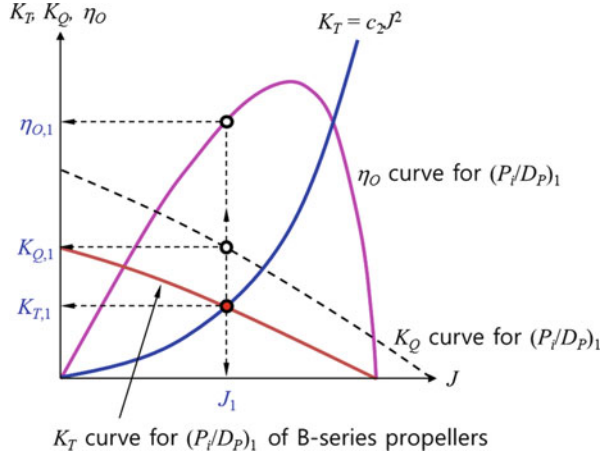
Substituting Eq. (7.21) into Eq. (7.20), it gives the following equation.

$$\begin{aligned} K_T &= \frac{R_T}{(1-t) \cdot \rho \cdot D_P^4} \cdot \left(\frac{J \cdot D_P}{V_A} \right)^2 \\ &= \frac{R_T}{(1-t) \cdot \rho \cdot D_P^2 \cdot V_A^2} J^2 = c_2 J^2 \end{aligned} \quad (7.22)$$

where, c_2 is defined as below and it can be calculated from the given data of Stage 2.

$$c_2 = \frac{R_T}{(1-t) \cdot \rho \cdot D_P^2 \cdot V_A^2}$$

Fig. 7.8 Calculation of the intersection point between K_T curves



Now, using POW diagram of the Wageningen B-series propellers for given values of A_E/A_O and z , we calculate the intersection points between K_T curves of the series propeller and quadratic curves of K_T for design propeller in Eq. (7.22).

As shown in Fig. 7.8, the POW diagram is given for specific values of A_E/A_O , z , and P_i/D_P . The quadratic curves of K_T in Eq. (7.22) is also plotted in this diagram. From two curves of K_T , the intersection point can be obtained. For the intersection point of $K_{T,1}$, the advance ratio (J_1), the K_Q coefficient ($K_{Q,1}$), and the propeller efficiency ($\eta_{O,1}$) at J_1 can be also found.

If we repeat this step for various pitch ratios, the advance ratio (J_x) which maximizes the propeller efficiency can be determined, as shown in Fig. 7.9.

From the pitch ratio which corresponds to the intersection point, the pitch can be obtained from the following equation.

$$P_{i,x} = (P_i/D_P)_x \cdot D_P \quad (7.23)$$

Using J_x and Eq. (7.21), the corresponding speed of propeller (same as the engine speed) can be calculated by the following equation.

$$n_x = \frac{V_A}{J_x \cdot D_P} \quad (7.24)$$

Now, using the torque equation in Eq. (7.18), the engine power can be calculated by the following equation.

$$P_x = 2\pi \cdot \rho \cdot n_x^3 \cdot D_P^5 \cdot K_{Q,x} \quad (7.25)$$

In the above equations, P_x and n_x are the engine power (P_E) and the engine speed (n_E) which correspond to the propeller having the maximum propeller efficiency and the pitch ($P_{i,x}$). These are solutions of Stage 2.

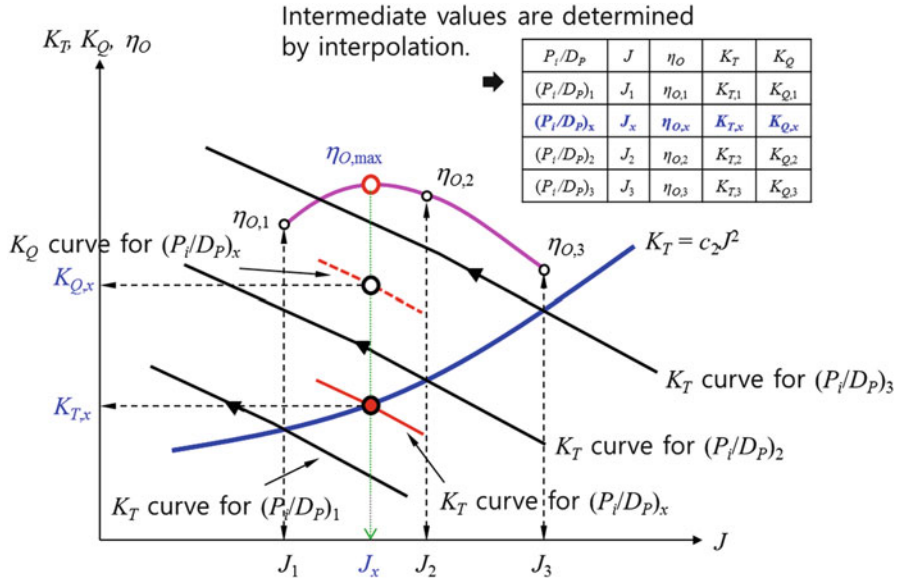


Fig. 7.9 Determination of the advance ratio for the maximum propeller efficiency

With the use of this engine power, BHP , NCR , and MCR of the design ship can be predicted again. At this time, η_R should be considered to change from the delivered power without the hull to the delivered power at the after body because the POW diagrams were generated from POW tests of propellers in open water.

$$BHP = \frac{DHP}{\eta_T} = \frac{P_x/\eta_R}{\eta_T} = \frac{P_x}{\eta_R \cdot \eta_T} \quad (7.26)$$

$$NCR = BHP \left(1 + \frac{\text{Sea Margin}}{100} \right) \quad (7.27)$$

$$MCR = \frac{NCR}{\text{Engine Margin}} \quad (7.28)$$

Also, the engine speed at NCR and MCR can be obtained from the following equations by using the relation between the propeller power and the propeller speed in Eq. (7.4) and by considering the match between main engine and propeller.

$$n_{NCR} = \sqrt[3]{\frac{NCR}{c_3}} \quad (7.29)$$

$$n_{MCR} = \sqrt[3]{\frac{MCR}{c_3}} \quad (7.30)$$

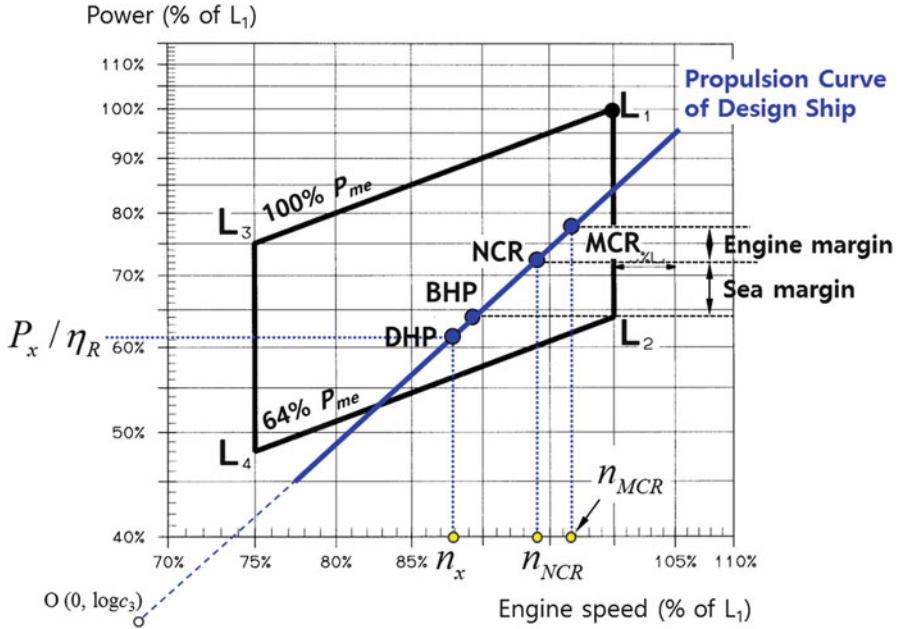


Fig. 7.10 Power prediction from the engine power and the engine speed determined at Stage 2

where,

$$c_3 = \frac{DHP}{n_x^3} = \frac{P_x/\eta_R}{n_x^3}$$

Figure 7.10 shows the result of power prediction according to the above equations.

Finally, if $NMCR$ of the main engine of design ship (initially, the main engine can be assumed as that of parent ship) cannot include NCR and MCR , including engine speeds at those powers, the main engine of design ship should be reselected as another one having greater power.

7.4.3 Stage 3: Determination of Optimal Principal Dimensions of Propeller and Maximum Ship Speed

At Stage 3, optimal principal dimensions of the propeller and the maximum speed of the ship will be determined. That is, with the engine power and the engine speed determined at Stage 2, principal dimensions of the propeller such as propeller diameter, propeller pitch, and EAR are determined. At this time, the number of blades is fixed as that of parent ship like in the problem of Stage 2.

At first, the **design point of propeller** should be determined as below. Figure 7.11 shows how to determine the design point of the propeller in the layout diagram of the main engine in logarithmic scales. In this diagram, we can define the load diagram as the power and speed limits for continuous as well as overload operation of an installed engine having an optimizing point and a specified MCR point according to the ship's specification. In this figure, Curve ① represents the **propulsion curve** with clean hull called '**light running propeller**'. Curve ② represents the propulsion curve with fouled hull and heavy weather called '**heavy running propeller**'. Curve ③ represents the torque and speed limit of the main engine at which an ample air supply is available for combustion and imposes a limitation on the maximum combination of torque and speed. This limit is proportional to the engine speed to the second ($\propto n_E^2$). Curve ④ represents the maximum level of mean effective pressure of the main engine for continuous operation. Curve ⑤ represents the maximum power for continuous operation of the main engine. Curve ⑥ represents the maximum speed of the main engine for continuous operation. The point M corresponds to the **specified MCR** (Maximum Continuous Rating or **Derated MCR**), the point S is *NCR* (Normal Continuous Rating), and the point B is *BHP*.

The point M can be any point within the layout diagram of the main engine. When choosing this point, we have to consider the propeller speed, the fuel consumption, and others together. Once the point M has been chosen the shaft and auxiliary equipment have been dimensioned accordingly, *the point M is now the maximum continuous rating (MCR)*. That is, *MCR* becomes the basis of all power and speed of the main engine.

The point G_1 is the intersection point of Curve ① with the power considering the sea margin and the point G_2 is the intersection point of Curve ① with the reduced power without taking into account the sea margin. One of these points can be used as the design point of propeller but normally the point G_1 is used as the design point.

Similar to Fig. 7.11, Fig. 7.12 shows the design point of the propeller in the layout diagram of the main engine in real scales. In this figure, Curve ②' represents the propulsion curve with the increased resistance as compared with Curve ②.

In this figure, the reason why one of the design points of propeller should be G_1 is as follows. If the propeller were designed at the point S, the propulsion curve would become Curve ②. When the resistance of ship increases with time, then the propulsion curve will move to the curve ②'. Thus, the propeller and engine match at the point S', at which the engine power is smaller than *NCR*. This means the propeller cannot absorb the engine power of *NCR*, which results in the reduction of ship speed. On the other hand, if the propeller was designed at the point G_1 , the propulsion curve would become Curve ①. When the resistance of ship increases with time, the propulsion curve will move to Curve ② so that the propeller can operate at the point S, at which the propeller can absorb the engine power of *NCR*. In summary, the **design point of the propeller** can be the point G_1 (*NCR*, n_{MCR}) or the point G_2 (*BHP*, n_{NCR}).

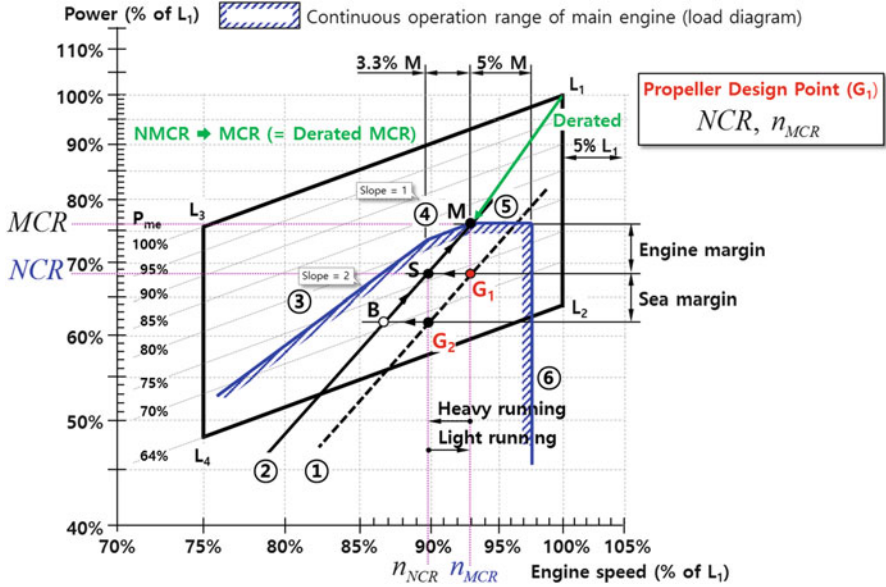


Fig. 7.11 Design point of the propeller in the layout diagram of the main engine in logarithmic scales

Now, the problem of Stage 3 for the determination of optimal principal dimensions of the propeller and the maximum speed of ship can be summarized in Table 7.3. If the point G_1 is used as the design point of the propeller, the engine power and the engine speed are NCR and n_{MCR} from Stage 2, as shown in Table 7.3.

In solving this problem, we can use three governing equations. The first one is that the propeller should absorb the torque delivered from the main engine, as shown in Eq. (7.18). The second one is that the propeller should produce the required thrust force at a given speed, as shown in Eq. (7.19). The last one an equation is about the non-cavitation criterion of the propeller. That is, this equation says that EAR (A_E/A_O) should be determined so that the cavitation does not occur. If EAR becomes smaller, the possibility of cavitation becomes higher although the propeller efficiency becomes higher. Thus, this equation gives the minimum requirement of EAR , as shown in Eq. (7.31) called **Keller's formula for minimum EAR**.

$$A_E/A_O \geq K + \frac{(1.3 + 0.3z) \cdot T}{D_P^2 \cdot (p_0 + \rho g h^* - p_v)} \quad (7.31)$$

where, K is 0.2 for ships with a single propeller and 0.1 for ships with two propellers (i.e., twin-screw ships). T is the thrust force of the propeller in kN. ($p_0 - p_v$) is 0.9947 kN/m² at sea water of 15 °C. h^* and h are the shaft immersion depth in m and the shaft center height (height from the baseline) in m, respectively.

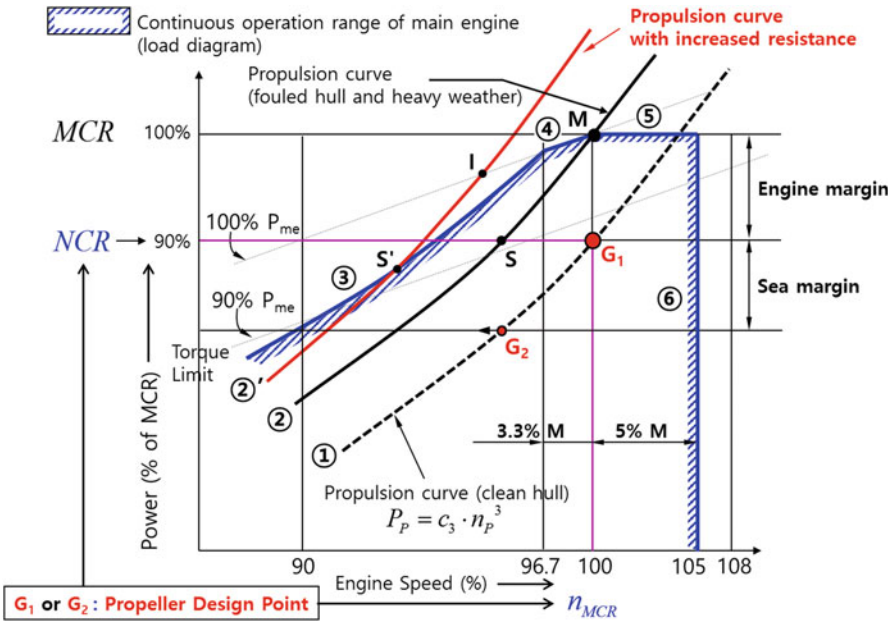


Fig. 7.12 Design point of the propeller in the layout diagram of the main engine in real scales

Table 7.3 Problem definition at Stage 3

	Item	Unit	Description
Given	P_E	kW	Delivered power to propeller from main engine (= NCR)
	n_E	1/s	Engine speed n_{MCR} (= Propeller speed, n_P)
	z	–	Number of blades
	R_T	kN	Resistance according to ship speed
Find	D_P	m	Propeller diameter
	P_i	m	Propeller pitch
	A_E/A_O	–	Expanded area ratio (EAR)
	$V_{\max}$	m/s	Maximum ship speed

Instead of this formula, **Burrill’s formula for minimum EAR** can be also used as the following equation.

$$A_E/A_O \geq \frac{F\left(\eta_O/(1/J)^2\right)}{\left(1 + 4.826(1/J)^2\right)^{0.375} \cdot (1.067 - 0.229P_i/D_P)} \tag{7.32}$$

where, F and B_P are calculated as below and DHP is in bhp (British Horse Power), V_A is in knots, and n_P is in rpm (1/min).

$$F = \frac{\eta_R \cdot B_P^2 \cdot V_A^{1.25}}{278.4(10.18 + h)^{0.625}}$$

$$B_P = n_P \cdot (DHP \cdot \eta_R)^{0.5} / V_A^{2.5}$$

The problem formulation for Stage 3 has four unknowns (D_P , P_i , A_E/A_O , and V) with two equalities (equations) and one inequality. Thus, this problem is a kind of an indeterminate problem since the number of unknowns is greater than the number of equations. If we assume the problem as an optimization problem by introducing an objective function, we can get the best solution which optimizes the objective function as in Stage 2.

The indeterminate problem can be solved by manually, as well. If the values of two unknowns (= the number of unknowns – the number of equations = $4 - 2 = 2$) are assumed, the indeterminate problem becomes a determinate problem and thus one solution can be obtained.

Here, we will solve this problem by assuming the values of A_E/A_O , and V . The detailed steps are given as below. *In the first step, we assume the value of A_E/A_O as a specific value. In the second step, we assume the value of V as a specific value, as well. In the third step, from Eq. (7.18) the torque coefficient (K_Q) can be expressed as a quintic function with respect to the advance ratio (J).*

That is, the torque equation in Eq. (7.18) can be rearranged as the following equation.

$$K_Q = \frac{P_E}{2\pi \cdot \rho \cdot n_P^3} \cdot \frac{1}{D_P^5} \quad (7.33)$$

The advance ratio can be expressed as the following equation.

$$D_P = \frac{V_A}{J \cdot n_P} \quad (7.34)$$

Substituting Eq. (7.34) into Eq. (7.33), it gives the following equation.

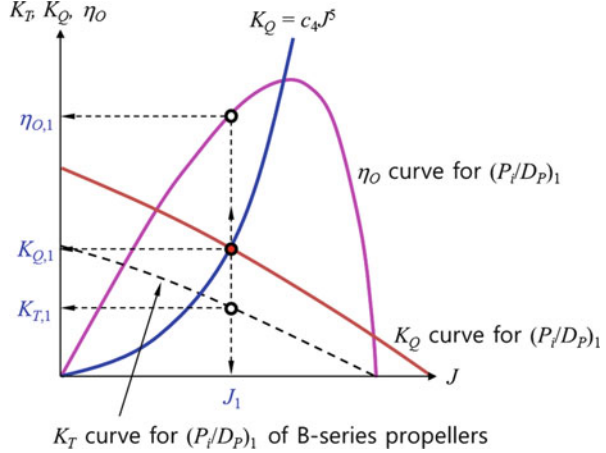
$$K_Q = \frac{P_E}{2\pi \cdot \rho \cdot n_P^3} \cdot \left(\frac{J \cdot n_P}{V_A} \right)^5$$

$$= \frac{P_E \cdot n_P^2}{2\pi \cdot \rho \cdot V_A^5} J^5 = c_4 J^5 \quad (7.35)$$

where, c_4 is defined as below and it can be calculated from the determined data at Stage 2.

$$c_4 = \frac{P_E \cdot n_P^2}{2\pi \cdot \rho \cdot V_A^5}$$

Fig. 7.13 Calculation of the intersection point between K_Q curves



In the fourth step, using POW diagram of the Wageningen B-series propellers for given values of A_E/A_O and z , we calculate the intersection points between K_Q curves in the POW diagram and quintic curves of K_Q in Eq. (7.35).

As shown in Fig. 7.13, the POW diagram is given for specific values of A_E/A_O , z , and P_i/D_P . The quintic curves of K_Q in Eq. (7.35) is also plotted in this diagram. From two curves of K_Q , the intersection point can be obtained. For the intersection point of $K_{Q,1}$, the advance ratio (J_1), the K_T coefficient ($K_{T,1}$), and the propeller efficiency ($\eta_{O,1}$) at J_1 can be also found.

If we repeat this step for various pitch ratios, the advance ratio (J_x) which maximizes the propeller efficiency can be determined, as shown in Fig. 7.14.

Using J_x and Eq. (7.34), the diameter of the propeller can be calculated by the following equation.

$$D_{P,x} = \frac{V_A}{J_x \cdot n_E} \quad (7.36)$$

From the pitch ratio which corresponds to the intersection point, the pitch can be obtained from the following equation.

$$P_{i,x} = (P_i/D_P)_x \cdot D_{P,x} \quad (7.37)$$

In the fifth step, the thrust force equation in Eq. (7.19) which means the relation between the ship and the propeller should be checked. Eq. (7.19) can be stated as the following equation.

$$\frac{R_T}{1-t} = \rho \cdot n_x^2 \cdot D_{P,x}^4 \cdot K_{T,x} \quad (7.38)$$

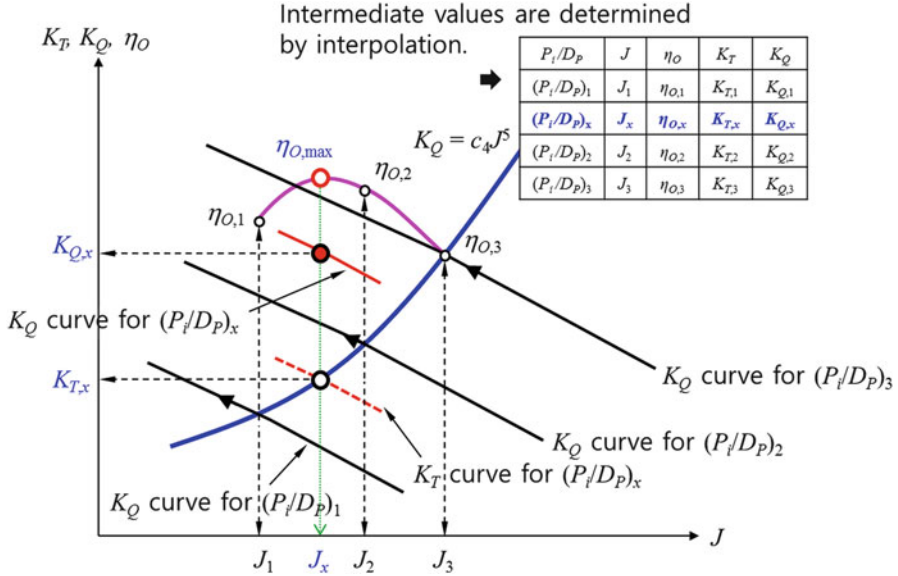


Fig. 7.14 Determination of the advance ratio for the maximum propeller efficiency

In this equation, total resistance (R_T) is a function of ship speed. *If this equation is not satisfied, we go to the second step and assume the ship speed with another value.* If this equation is satisfied, the ship speed becomes the maximum ship speed ($V_{\max}$) with the determined principal dimensions of the propeller and we go to the next step. *In the last step, Eq. (7.31) or Eq. (7.32) which means the non-cavitation criterion of propeller should be checked.* If Eq. (7.31) is used here, Eq. (7.39) should be checked.

$$A_E/A_O \geq K + \frac{(1.3 + 0.3z) \cdot T_x}{D_{P,x}^2 \cdot (p_0 + \rho g h^* - p_v)} \quad (7.39)$$

where, T_x can be obtained from the following equation.

$$T_x = \rho \cdot n_x^2 \cdot D_{P,x}^4 \cdot K_{T,x} \quad \text{or} \quad T_x = \frac{R_T}{1 - t} \quad (7.40)$$

If this inequality is not satisfied, we go to the first step and assume the EAR with another value. If this inequality is satisfied, the values of the principal dimensions of the propeller and the maximum ship speed will be the best solution of this problem. The steps for finding the best solution for this problem can be represented in Fig. 7.15.

Fig. 7.15 Overall steps for solving the problem of Stage 3

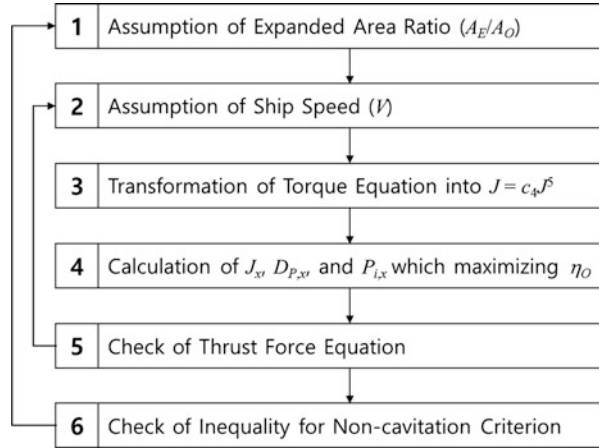


Table 7.4 Problem definition in Stage 4

	Item	Unit	Description
Given	D_P	m	Propeller diameter
	P_i	m	Propeller pitch
	A_E/A_O	–	Expanded area ratio (EAR)
	z	–	Number of blades
	$V_{\max}$	m/s	Maximum ship speed
	R_T	kN	Resistance according to ship speed
Find	P_E	kW	Delivered power to propeller from main engine (= NCR)
	n_E	1/s	Engine speed n_{MCR} (= Propeller speed, n_P)

7.4.4 Stage 4: Determination of Engine Power and Speed Using the Determined Propeller Dimensions

At Stage 4, the engine power and the engine speed are re-determined by using the determined optimal dimensions of the propeller in the previous step. That is, this stage is *similar to Stage 2*. The problem for determining them can be summarized in Table 7.4.

When we solve this problem, we use two governing equations which were already described above. The first one is the torque equation, as shown in Eq. (7.18). The second one is the thrust force equation, as shown in Eq. (7.19).

The problem of Stage 4 has two unknowns (P_E and n_E) and two equations. Thus, this problem is a kind of a determinate problem since the number of unknowns is equal to the number of equations. The determinate problem has a unique solution if the equations are independent of each other. The problem can be stated as below.

Find P_E and n_E

which

Satisfy

$$\frac{P_E}{2\pi \cdot n_P} = \rho \cdot n_P^2 \cdot D_P^5 \cdot K_Q$$

$$\frac{R_T}{1-t} = \rho \cdot n_P^2 \cdot D_P^4 \cdot K_T$$

First, from Eq. (7.19) the thrust coefficient (K_T) can be expressed as a quadratic function with respect to the advance ratio (J). As described in the problem of Stage 2, the thrust force equation in Eq. (7.19) can be rearranged as Eq. (7.41).

$$K_T = c_2 J^2 \quad (7.41)$$

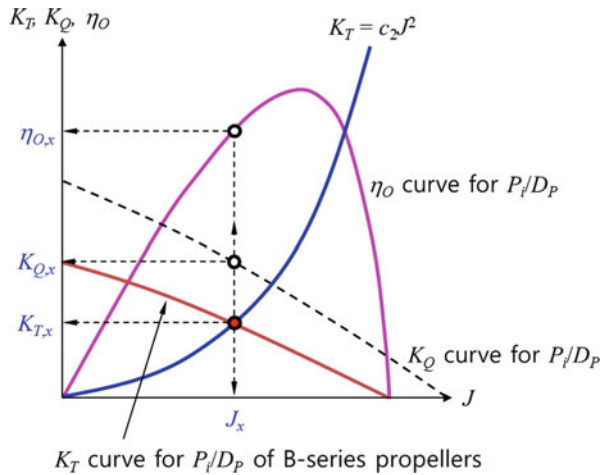
where, c_2 is defined as below and it can be calculated from the given data of Stage 4.

$$c_2 = \frac{R_T}{(1-t) \cdot \rho \cdot D_P^2 \cdot V_A^2}$$

Now, using POW diagram of the Wageningen B-series propellers for given values of A_E/A_O , z , and P_i/D_P , we calculate the intersection point between a K_T curve in the POW diagram and a quadratic curve of K_T in Eq. (7.41).

As shown in Fig. 7.16, the POW diagram is given for specific values of P_i/D_P , A_E/A_O , and z . The quadratic curve of K_T in Eq. (7.41) is also plotted in this diagram. From two curves of K_T , the intersection point can be obtained. For the intersection

Fig. 7.16 Calculation of the intersection point between K_T curves



point of $K_{T,x}$, the advance ratio (J_x), the K_Q coefficient ($K_{Q,x}$), and the propeller efficiency ($\eta_{O,x}$) at J_x can be also found.

Using J_x and Eq. (7.21), the corresponding speed of propeller (same as the engine speed) can be obtained from the following equation.

$$n_x = \frac{V_A}{J_x \cdot D_P} \quad (7.42)$$

Now, using the torque equation in Eq. (7.18), the engine power can be calculated by the following equation.

$$P_x = 2\pi \cdot \rho \cdot n_x^3 \cdot D_P^5 \cdot K_{Q,x} \quad (7.43)$$

In the above equations, P_x and n_x are the engine power of NCR and the engine speed at MCR (n_{MCR}) which correspond to the design point of the propeller at Stage 3. These are the solution for Stage 4 (P_E and n_E , respectively). At this time, we check NCR and n_{MCR} are same as those values at Stage 3 which were bases for the determination of optimal principal dimensions of the propeller. If they are not same to each other, we go to Stage 3 and repeat to solve the problem at Stage 3 until they converge. If all is done, the solution will be the final principal dimensions of the propeller having the maximum propeller efficiency.

7.4.5 Stage 5: Generation of Speed-Power Curve

At Stage 5, we determine the required power and the engine speed for a given speed of the ship. Furthermore, we determine the ship speed for a given power and engine speed. That is, we generate the speed-power curve. First, we repeat the steps with the optimal propeller at Stage 4 for various ship speeds. Then, calculate c_2 , J , n_E ($= n_P$), and η_O for each ship speed. In addition, EHP, DHP, and BHP can be calculated by using Eqs. (7.44), (7.45) and (7.46).

$$EHP = R_T \cdot V \quad (7.44)$$

$$DHP = \frac{EHP}{\eta_D} = \frac{EHP}{\eta_O \cdot \eta_H \cdot \eta_R} \quad (7.45)$$

$$BHP = \frac{DHP}{\eta_T} \quad (7.46)$$

From these calculations, we can get a table of the power prediction for different ships speeds. Table 7.5 shows an example of the power prediction for different ship speeds ranging from 12.5 to 14.5 knots with a step of 0.5 knots.

In this table, PS is one of the units for the power, that is, German horse power. 1 PS = 75 kgf·m/s = 0.73575 kW.

Table 7.5 Example of the power prediction for different ship speeds

Speed (knot)	EHP (PS)	w	t	BHP (PS)	$c_2 = K_T/J^2$	J	n_E or n_P (rpm)	η_O
12.5	1,686	0.381	0.224	2,867	1.374	0.355	202	0.470
13.0	1,965	0.380	0.223	3,367	1.418	0.352	212	0.465
13.5	2,240	0.379	0.221	3,844	1.435	0.351	221	0.463
14.0	2,536	0.377	0.219	4,376	1.443	0.348	232	0.460
14.5	2,898	0.375	0.216	5,020	1.470	0.345	243	0.457

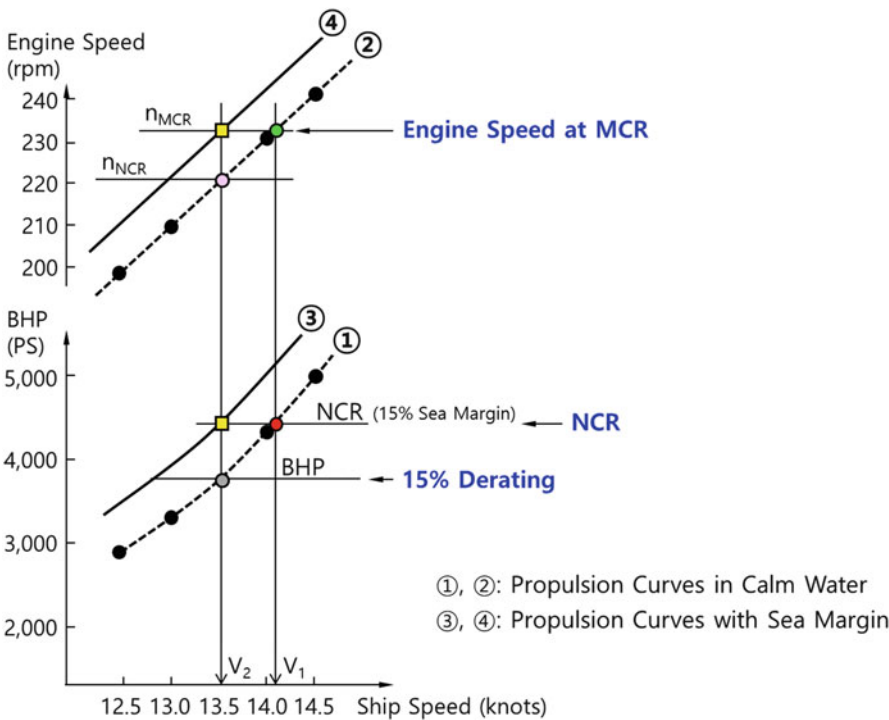


Fig. 7.17 Generation of speed and power curves considering sea margin

With the data in this table, we can mark points of the engine power (*BHP*) and the engine speed (n_E) according to the ship speeds, as shown in Fig. 7.17. By connecting the points (black circle) for *BHP* and n_E , the **propulsion curves** (also called the **speed-power curves**) in the calm water (① and ②, dashed line) can be made.

Now, we draw a horizontal line corresponding to *NCR* (design point of the propeller) and get an intersection point (red circle) with Curve ① of the engine power at the ship speed V_1 . On a vertical line corresponding to that speed, if we get an intersection point (green circle) with Curve ② of the engine speed, the point corresponds to n_{MCR} . Now, we can draw a horizontal line corresponding to *BHP*

without considering sea margin (i.e., calm sea) and get an intersection point (gray circle) with Curve ① of the engine power at the ship speed V_2 . On a vertical line corresponding to that speed, we get an intersection point (pink circle) with Curve ② of the engine speed, and intersection points (yellow boxes) with horizontal lines corresponding to NCR and the n_{MCR} . If we move Curves ① and ② so that they pass through the yellow boxes, we can get new Curves ③ and ④. These curves mean **propulsion curves with sea margin**. Finally, with these curves, we can get the ship speed and the engine speed (or the propeller speed) under the given engine power. That is, if NCR is given as the engine power, the ship speed (service speed) will be 13.5 knots in this example.

7.5 Relation Between Propeller Speed, Diameter, and Efficiency

The relation between the propeller speed (same as the engine speed in case of no reduction gears), diameter, and the propeller efficiency is discussed below.

We assume that the optimum advance ratio (J) in Eq. (7.8) is constant in the optimum range. Under this assumption, if the propeller speed (n) decreases, the propeller diameter (D_P) will increase. Then, the torque coefficient (K_Q) in Eq. (7.11) becomes smaller than the thrust coefficient (K_T) in Eq. (7.9). Finally, the propeller efficiency (η_O) in Eq. (7.13) increases.

From this discussion, if the propeller speed decreases, the optimum diameter of the propeller becomes larger, and the propeller efficiency increases. On the other hand, if the propeller speed increases, the optimum diameter of the propeller becomes small, and the propeller efficiency decreases.

7.6 Computational Guide

In this section, a computational guide to determine the principal dimensions of the propeller is briefly described.

7.6.1 Stage 1

1. Problem definition

The problem for this stage can be defined as below.

- (a) Given: propeller diameter ($D_{P,P}$), MCR (MCR_P), and the engine speed at MCR ($n_{MCR,P}$) of the parent ship